

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0867

Parkin 138 kV
May 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0867. The Interconnection Customer proposes a 238.9 MW wind facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on September 28, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 302 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,947,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Oak Creek – Carmel 146Th St 138 kV
Contingency: Bear – Clover 345 kV L
Loading: 176.7% (Rating: 200 MVA)
DFAX: 6.4%
Pre-project loading: 169.0%
2. Facility: Ceiba Baja 7012 – Carmel 146Th St 138 kV
Contingency: Clovr – Bear 345 kV
Loading: 174.8% (Rating: 204 MVA)
DFAX: 6.4%
Pre-project loading: 167.3%
3. Facility: Ceiba Baja 7012 – Tangy 138 kV L
Contingency: Bear – Clover 345 kV Line
Loading: 160.3% (Rating: 229 MVA)
DFAX: 6.4%
Pre-project loading: 153.6%
4. Facility: S Colton – Flowood 138 kV
Contingency: Parkin – Candler's Mtn 138 kV Line
Loading: 140.7% (Rating: 242 MVA)
DFAX: 40.6%
Pre-project loading: 100.6%
5. Facility: Flowood – Unity 138 kV Line
Contingency: S Creek Substation – Wythe 138 kV line
Loading: 125.0% (Rating: 345 MVA)
DFAX: 44.5%
Pre-project loading: 94.1%

6. Facility: Kerr Dam – Mecklenburg Power Sta Substation 345 kV
Contingency: Bear Substation – Mecklenburg Power Station 345 kV line
Loading: 123.1% (Rating: 1,198 MVA)
DFAX: 10.6%
Pre-project loading: 121.0%
7. Facility: Unity – Wythe 138 kV Line
Contingency: S Crk – Wythe 138 kV Line
Loading: 122.6% (Rating: 368 MVA)
DFAX: 44.5%
Pre-project loading: 93.7%
8. Facility: Wythe – S Creek 138 kV
Contingency: Candler's Mtn – Parkin 138 kV
Loading: 122.3% (Rating: 517 MVA)
DFAX: 46.1%
Pre-project loading: 101.0%
9. Facility: New London – Barksdale 138 kV
Contingency: Opossum Crk – Parkin Substation 138 kV
Loading: 104.5% (Rating: 317 MVA)
DFAX: 7.0%
Pre-project loading: 99.2%
10. Facility: Intel – S Crk 138 kV
Contingency: Aibonito Pds – South Crk 138 kV
Loading: 104.4% (Rating: 341 MVA)
DFAX: 17.0%
Pre-project loading: 92.4%
11. Facility: Boonsboro – Amargo 138 kV
Contingency: Parkin – Opossum Crk 138 kV L
Loading: 102.3% (Rating: 1,001 MVA)
DFAX: 100.0%
Pre-project loading: 78.5%
12. Facility: Brookville – Amargo 138 kV Line
Contingency: Opossum Creek – Parkin 138 kV
Loading: 102.0% (Rating: 1,025 MVA)
DFAX: 100.0%
Pre-project loading: 78.7%
13. Facility: S Lynchburg – Brookville 138 kV L
Contingency: Parkin Substation – Opossum Crk 138 kV Line
Loading: 101.5% (Rating: 1,064 MVA)
DFAX: 100.0%
Pre-project loading: 79.0%
14. Facility: Candler's Mountain – S Lynchburg 138 kV
Contingency: Parkin – Opossum Crk 138 kV
Loading: 101.3% (Rating: 1,075 MVA)
DFAX: 100.0%
Pre-project loading: 79.1%
15. Facility: Central Tie – Wythe 138 kV
Contingency: Parin – Candler's Mtn 138 kV Line
Loading: 101.3% (Rating: 1,076 MVA)
DFAX: 100.0%
Pre-project loading: 79.1%
16. Facility: Central Tie Substation – Opossum Creek 138 kV
Contingency: Parkin – Candler's Mtn 138 kV

Loading: 101.1% (Rating: 1,088 MVA)

DFAX: 100.0%

Pre-project loading: 79.2%

17. Facility: Candler's Mountain – Parkin 138 kV L

Contingency: Opossum Crk – Parkin 138 kV

Loading: 101.0% (Rating: 1,099 MVA)

DFAX: 100.0%

Pre-project loading: 79.3%

18. Facility: Parkin – Opossum Creek 138 kV

Contingency: Candler's Mountain – Parkin Substation 138 kV Line

Loading: 101.0% (Rating: 1,099 MVA)

DFAX: 100.0%

Pre-project loading: 79.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carmel 146Th St – Oak Crk 138 kV	Rebuild line with higher-capacity conductor	\$6,854,000	18% (\$1,235,000)
Ceiba Baja 7012 – Carmel 146Th St 138 kV	Rebuild line with higher-capacity conductor	\$12,963,000	18% (\$2,336,000)
Ceiba Baja 7012 – Tangy 138 kV	Rebuild line with higher-capacity conductor	\$6,435,000	18% (\$1,160,000)
Flowood Substation – South Colton 138 kV	Rebuild line with higher-capacity conductor	\$4,261,000	21% (\$898,000)
Flowood – Unity 138 kV	Rebuild line with higher-capacity conductor	\$6,790,000	22% (\$1,458,000)
Kerr Dam – Mecklenburg Power Sta Substation 345 kV	Rebuild line with higher-capacity conductor	\$39,266,000	6% (\$2,364,000)
Wythe – Unity 138 kV	Rebuild line with higher-capacity conductor	\$9,285,000	22% (\$1,994,000)
Wythe – S Creek 138 kV	Rebuild line with higher-capacity conductor	\$11,010,000	21% (\$2,339,000)
New London – Barksdale 138 kV	Reconductor line	\$5,312,000	17% (\$909,000)
Intel – S Crk 138 kV	Reconductor line	\$3,721,000	17% (\$640,000)
Boonsboro – Amargo 138 kV	Reconductor line	\$1,694,000	22% (\$367,000)
Brookville – Amargo 138 kV	Reconductor line	\$225,000	22% (\$49,000)
S Lynchburg – Brookville 138 kV L	Reconductor line	\$1,081,000	22% (\$234,000)
South Lynchburg – Candler's Mountain 138 kV	Reconductor line	\$1,295,000	22% (\$280,000)
Central Tie – Wythe 138 kV	Reconductor line	\$784,000	22% (\$170,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Central Tie Substation – Opossum Creek 138 kV	Reconductor line	\$622,000	22% (\$135,000)
Candlers Mountain – Parkin 138 kV L	Reconductor line	\$361,000	22% (\$78,000)
Parkin – Opossum Crk 138 kV	Reconductor line	\$1,388,000	22% (\$301,000)
	Total allocated to project		\$16,947,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.